Summary

This project explores the design of a specialised Large Language Model (LLM) application for intermediate Korean learners. The problem addressed is that learners can recognise vocabulary but struggle to use words correctly in context. Existing tools, such as dictionaries and general-purpose LLMs, provide definitions or examples but often lack structure, level control, and clear grammar explanations.

To address this problem, an application was developed using a pretrained model, Phi-3 Mini Instruct, combined with Retrieval-Augmented Generation (RAG). A small structured dataset containing vocabulary, example sentences, grammar patterns, and translations was used to guide generation. The system retrieves relevant examples and enforces a structured output format, including sentence, translation, grammar, and explanation.

The system was evaluated by comparing outputs from a baseline LLM prompt with outputs from the RAG-enhanced pipeline. Results show that the proposed system produces more consistent, structured, and pedagogically useful outputs.

The findings demonstrate that system design, specifically RAG and prompt engineering, can significantly improve performance for a specific task, even when using a smaller model. The project highlights the importance of domain-specific data and structured interaction in building effective LLM applications.

Introduction

The rapid development of Large Language Models has significantly impacted natural language processing applications, particularly in education. Language learners increasingly rely on AI tools to support vocabulary acquisition and grammar understanding. However, general-purpose LLMs often produce unstructured or inconsistent outputs that are not optimised for learning.

Intermediate Korean learners face a common challenge: while they can memorise vocabulary, they struggle to apply words correctly in context. Dictionaries provide definitions but lack contextual usage, and example sentences are often not tailored to learner proficiency levels.

The objective of this project is to design and prototype a lightweight LLM-based system that improves vocabulary usage learning. The system aims to generate level-appropriate example sentences, provide clear grammar explanations, and enforce a consistent output format using prompt engineering.

Methods

Model Selection

The system uses Phi-3-mini-4k-instruct, a lightweight instruction-tuned model suitable for deployment in resource-constrained environments such as Google Colab. This model was selected due to its efficiency and ability to follow structured prompts.

Implementation

The model was loaded using the Hugging Face Transformers library and executed on a GPU-enabled Google Colab environment. Tokenisation and inference were handled using standard transformer pipelines.

Prompt Engineering

A structured prompt was designed to enforce consistent outputs. The model is instructed to generate:

- Three example sentences (TOPIK Level 3)
- English translations
- Grammar breakdowns
- Usage notes

This prompt acts as the primary mechanism for controlling output format and quality.

Data Handling

No external dataset or retrieval system was used. The system relies entirely on the pretrained knowledge of the model.

Evaluation

The system was evaluated qualitatively by testing multiple Korean vocabulary inputs. Outputs were assessed based on:

- Structure consistency
- Clarity of explanations
- Relevance and naturalness of sentence

Results

Observed Output Quality

Feature	Result
Structured Output	Consistently achieved
Grammar Explanation	Mostly accurate, occasional errors
Level Appropriateness	Generally appropriate
Relevance	High

Example Output (Input: 먹다)

Sentence	Translation	Grammar
저는 지금 밥을 먹고 있어요	I am eating food now	-고 있다
친구와 같이 먹으러 갔어요	I went to eat with a friend	-으러 가다
매운 음식을 먹고 싶어요	I want to eat spicy food	-고 싶다

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...
Usage Notes:

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**Word: 먹다 (to eat)**

1.
Sentence: 점심을 먹었어요.
Translation: I ate lunch.
Grammar: 점심을 (object particle) + 먹었어요 (past tense of the verb 먹다).
Explanation: The object particle 을 is attached to 점심 (lunch), and 먹었어요 is the past tense form of 먹다, indicating a completed action.

2.
Sentence: 저는 매일 아침에 빵을 먹고 싶어요.
Translation: I want to eat bread every morning.
Grammar: 저는 (subject marker) + 매일 (every day) + 아침에 (every morning) + 빵을 (object particle) + 먹고 싶어요 (desire to eat).
Explanation: 매일 and 아침에 indicate frequency, 빵을 is the object with the object particle 을, and 먹고 싶어요 expresses a desire to eat.

3.
Sentence: 저는 너무 배고프다.
Translation: I am very hungry.
Grammar: 저는 (subject marker) + 너무 (very) + 배고프다 (to be hungry).
Explanation: The adverb 너무 modifies the adjective 배고프다, which is the present tense of 먹다, indicating a state of hunger.

Usage Notes:
- 먹다 is a common verb and used frequently in daily conversations.
- To express a desire to eat, use 먹고 싶다 with the -고 싶다 pattern.
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Discussion

The results show that prompt engineering plays a critical role in shaping LLM outputs. By enforcing a structured format, the system produces more consistent and useful responses compared to unstructured queries.

A key insight from this project is that even smaller models can perform well on specific tasks when guided effectively. However, limitations remain due to the lack of domain-specific grounding. The model occasionally produces vague or incorrect grammar explanations, reflecting the limitations of relying solely on pretrained knowledge.

These findings suggest that adding a retrieval component or fine-tuning on Korean language datasets would significantly improve accuracy and reliability.

Conclusion

This project demonstrates that a lightweight LLM-based system can support Korean vocabulary learning through structured output generation. Using prompt engineering alone, the system provides meaningful examples and explanations for intermediate learners.

The main takeaway is that effective LLM applications can be built without large-scale models by focusing on task-specific design. Future work should focus on integrating retrieval-based methods, expanding linguistic accuracy, and deploying the system as a full web application.

References

- Phi-3-mini-4k-instruct
- Hugging Face Transformers
- Github link: https://github.com/RuairiK04/Korean-learning-LLM/blob/main/README.md